COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 010-CPE21S2 - Data Structures and Algorithms	**SECTION:**	CPE21S2
DATE PERFORMED:	26/08/2026	**DATE SUBMITTED:**	26/08/2026
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Prelim Laboratory Exam Set D

1. PROCEDURE OUTPUT

STACK/ARRAY

step 1 Start
step 2 push (1)
step 3 push (2)
step 4 push (3)
step 5 push (2)
step 6 push (1)
step 8 initialize the value of the top of the original stack to the COPY Stack and making use of the FIFO rule it follows
step 9 we check for if its symmetric by looking into the sequential order it was put on if all values are = then return 1 if not 0
step 10 print is the Stack Symmetric?
step 11 std::cout<<"Is stack in symmetric" std::boolalpha
step 12 display the return value

Analysis: (For stack) In order for us to make this program we would need to create two STACK one for the original where the inputs will be stored and another for where the value of the original stack will be copied from top to bottom making it so that after every value is stored inside the COPY STACK they would immediately be in reversed order of the ORIGINAL STACK and we will just have to check if each of its sequential values are equal to each other if they are then return 1 if not then 0

QUEUE/LINKEDLIST

step 1 myQ.push('1');
step 2 myQ.push('1');
step 3 myQ.push('3');
step 4 myQ.push('2');
step 5 myQ.push('1');
step 9 Create another temporary QUEQUE that we will use to copy the values of the QUEUE from top to bottom to top since it follows the FIFO rule.
step 10 Check if the every value in the temporary QUEUE is = to the original QUEUE making sure the conditions of the value is set to its sequential order.
Step 11 if every value QUEUE == temporary QUEUE then return 1 if not 0
step 12 print is the Queue Symmetric?
step 13 std::boolalpha so that it will display the return value in true or false
step 14 display the return value

Analysis:(For queue) we would need to have a function that returns the elements in reverse order of the original sequence it was added so that we will be able to check its symmetry between the original stack and the copied stack if the all the values inside the stack of the same sequence are equal return 1 if not 0;

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

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